Comparative Research on GPU, its Operating Systems, and Application in AWS Cloud Platforms

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Module Title: Computer Architecture and Networks

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1. Introduction

The contemporary workloads in computing in fields like artificial intelligence, machine learning, and scientific computing require high-performance computing hardware. Graphics Processing Bundles, or GPUs, are becoming more significant in such settings because of their parallel processing capabilities, where inefficient tasks by the normal CPU are accelerated on the GPUs. Since they are specific to large amounts of concurrent work, the GPUs are more suitable in graphics processing, training neural networks, and processing data. Besides hardware, the GPU operating system commands are also important in load balancing, controlling, and tracking the performance of GPUs, as this is significant in the process of deploying the workloads in cloud computing.

This report explores the application of GPU operating systems in the local and cloud environments. Task 1 examines the techniques of using GPU CLI commands within a system based on Windows with an Intel HD Graphics 520 card. Task 2 illustrates the implementation and setup of an AWS EC2 instance to deploy a workload to the cloud and details the steps to follow during the setup (Tattu, 2024). Task 3 will assess the level of effectiveness of processing GPU commands together with cloud platforms, whereas Task 4 will be a final statement of conclusions based on the outcomes of the study. This report will address the learning outcomes of the CN7029 module as well as a reflective assessment of technical competence in both GPU systems and cloud computing through a mix of practical instructions and screenshots as well as assessment.

2. Task 1: Comparative Research on GPU Operating Systems

[redacted]

Administrator: Windows PowerShell

```
PS C:\Windows\system32> # u2869613 - Task 1: List all GPUs installed in the system
>> Get-WmiObject Win32_VideoController | Select-Object Name, DriverVersion, VideoProcessor
>>

Name                    DriverVersion VideoProcessor
----                    ------------- --------------
Intel(R) HD Graphics 520 30.0.101.1338 Intel(R) HD Graphics Family


PS C:\Windows\system32>
```

```
PS C:\Windows\system32> # u2869613 - Task 1: Show GPU utilization and memory usage
>> Get-Counter "\GPU Engine(*)\Utilization Percentage"
>>

Timestamp                 CounterSamples
---------                 --------------
1/2/2026 2:29:02 AM       \\desktop-qa274h7\gpu
                          engine(pid_10028_luid_0x00000000_0x0000c3de_phys_0_eng_0_engtype_3d)\utilization percentage :
                          0

                          \\desktop-qa274h7\gpu engine(pid_10028_luid_0x00000000_0x0000c3de_phys_0_eng_10_engtype_gdi
                          render)\utilization percentage :
                          0

                          \\desktop-qa274h7\gpu
                          engine(pid_10028_luid_0x00000000_0x0000c3de_phys_0_eng_11_engtype_videoprocessing)\utilization
                          percentage :
                          0

                          \\desktop-qa274h7\gpu
                          engine(pid_10028_luid_0x00000000_0x0000c3de_phys_0_eng_12_engtype_)\utilization percentage :
                          0

                          \\desktop-qa274h7\gpu
                          engine(pid_10028_luid_0x00000000_0x0000c3de_phys_0_eng_1_engtype_videodecode)\utilization
                          percentage :
                          0

                          \\desktop-qa274h7\gpu
                          engine(pid_10028_luid_0x00000000_0x0000c3de_phys_0_eng_2_engtype_copy)\utilization percentage :
                          0

                          \\desktop-qa274h7\gpu
                          engine(pid_10028_luid_0x00000000_0x0000c3de_phys_0_eng_3_engtype_videoprocessing)\utilization
                          percentage :
                          0
```

```
PS C:\Windows\system32> # u2869613 - Task 1: Show GPU memory usage
>> Get-Counter "\GPU Adapter Memory(*)\Dedicated Usage"
>>

Timestamp                 CounterSamples
---------                 --------------
1/2/2026 2:32:37 AM       \\desktop-qa274h7\gpu adapter memory(luid_0x00000000_0x0000c3de_phys_0)\dedicated usage
                          0

                          \\desktop-qa274h7\gpu adapter memory(luid_0x00000000_0x0000c7bd_phys_0)\dedicated usage
                          0



PS C:\Windows\system32>
```

[redacted]

[redacted]

[redacted]

```
PS C:\Windows\system32> # u2869613 - Task 1: Show GPU driver details and status
>> Get-WmiObject Win32_VideoController | Select-Object Name, DriverVersion, Status, VideoModeDescription
>>

Name                     DriverVersion Status VideoModeDescription
----                     ------------- ------ --------------------
Intel(R) HD Graphics 520 30.0.101.1338 OK     2560 x 1440 x 4294967296 colors


PS C:\Windows\system32>
```

[redacted]

[redacted]

```
PS C:\Windows\system32> # u2869613 - Task 1: Show GPU temperature
>> Get-WmiObject MSAcpi_ThermalZoneTemperature -Namespace "root/wmi" |
>> Select-Object InstanceName, CurrentTemperature
>>

InstanceName            CurrentTemperature
------------            ------------------
ACPI\ThermalZone\THM__0               2982
```

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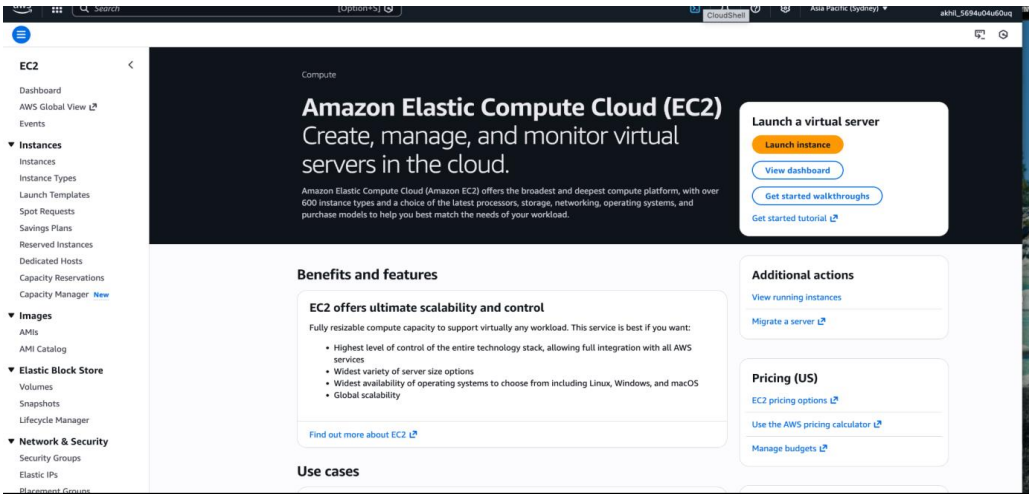

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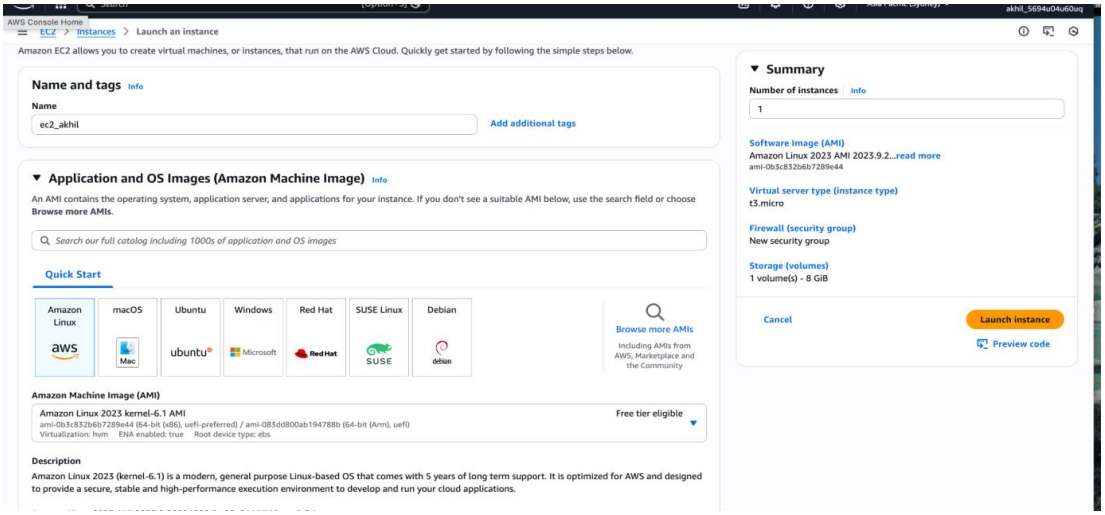

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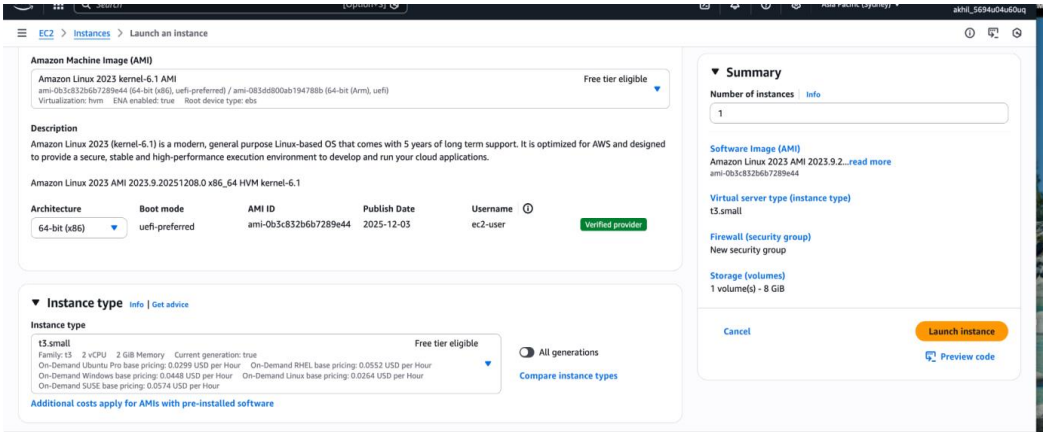

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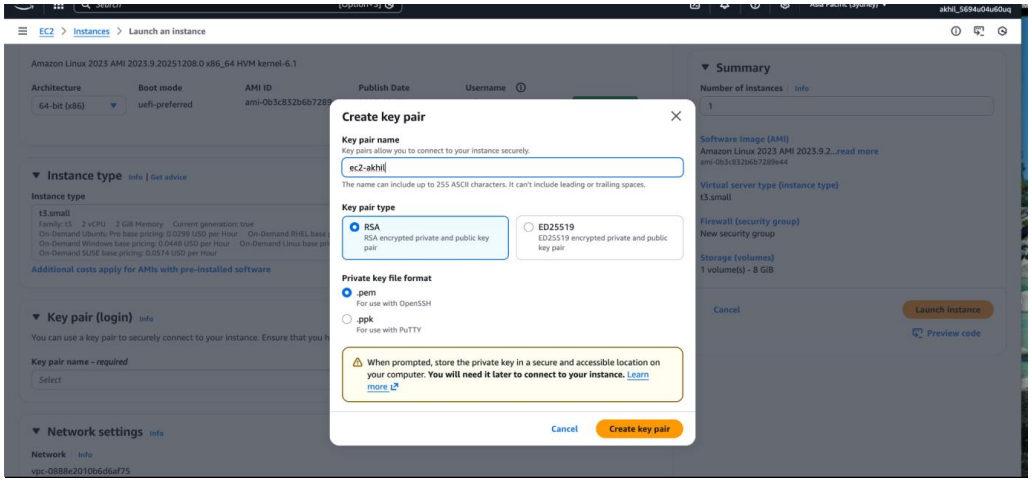

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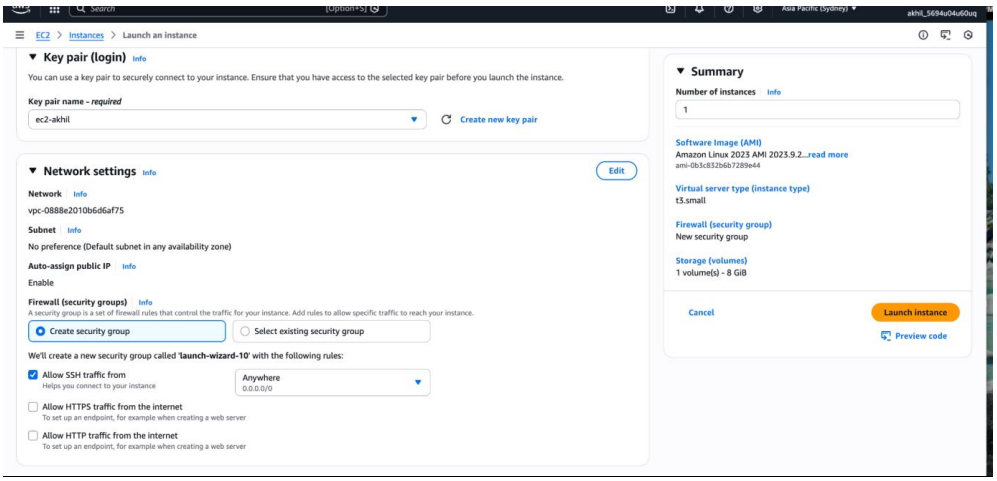

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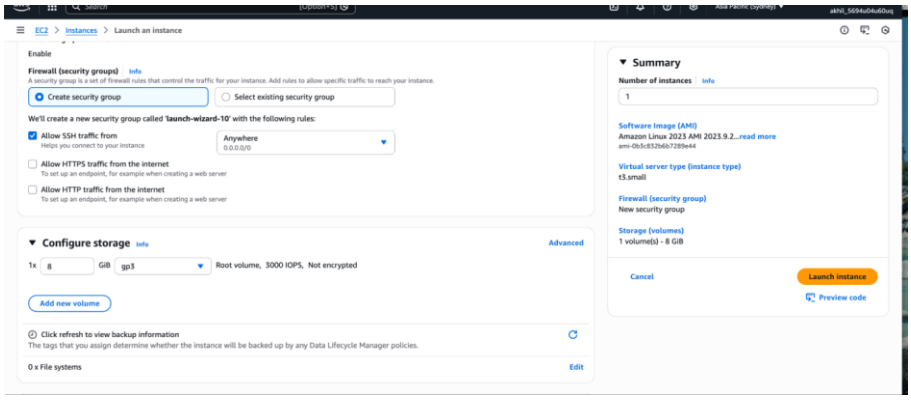

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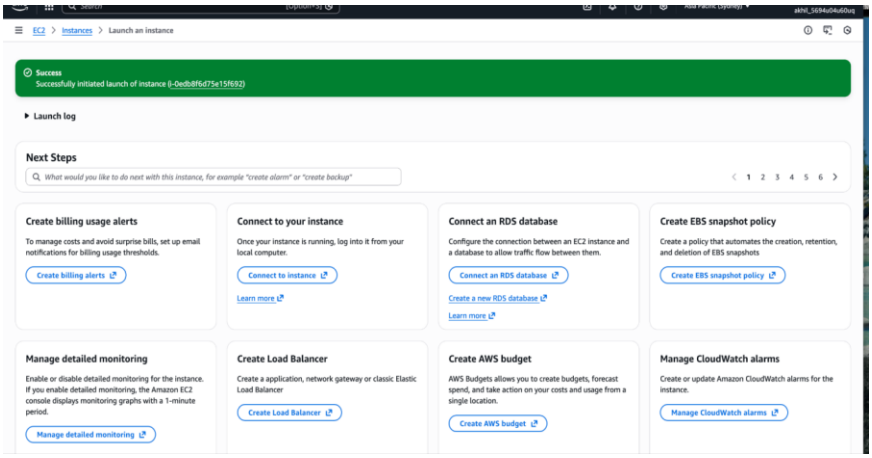

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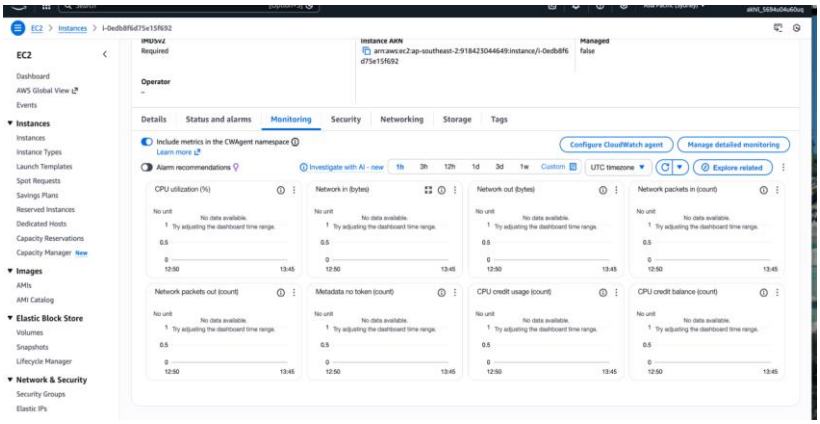

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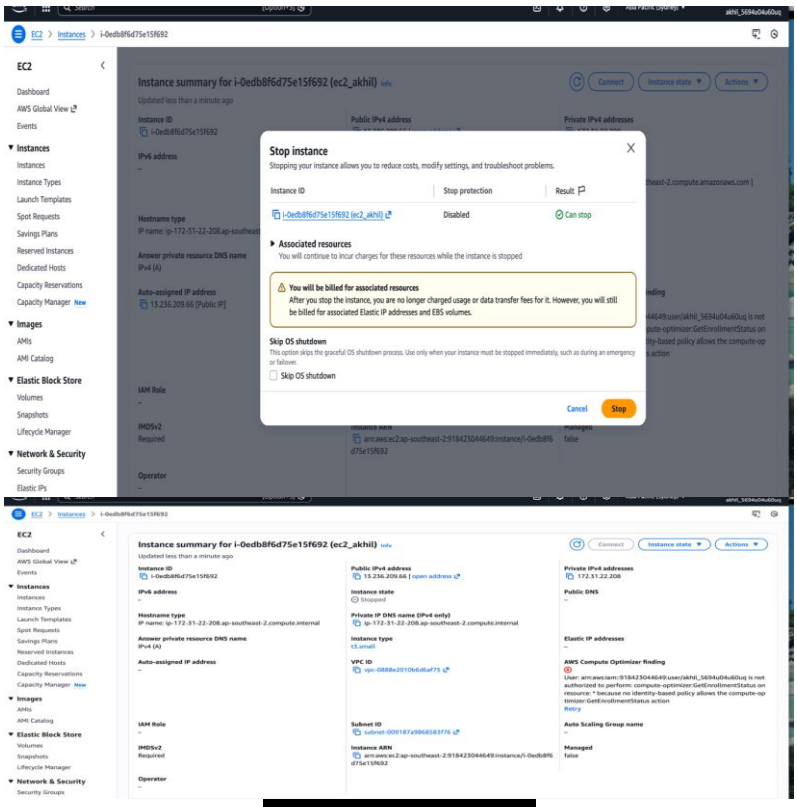

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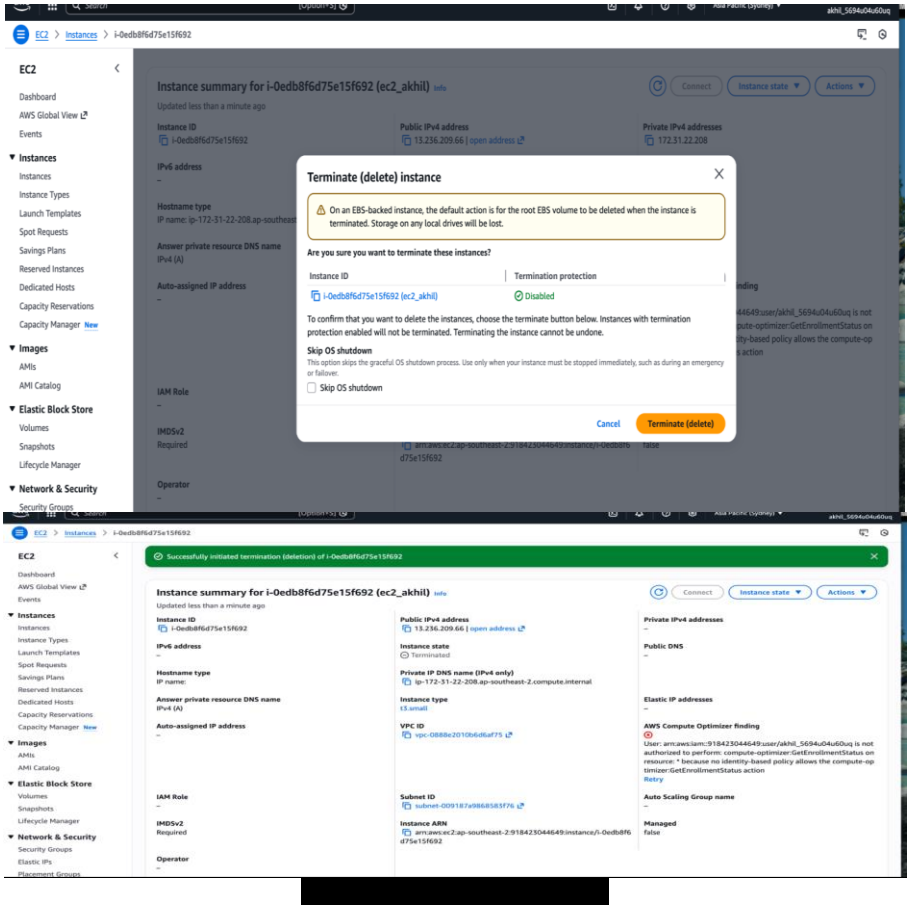

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